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Monday 7:17 AM

The following polynomial from C^3 to C^3 was just announced as a counterexample to the jacobian conjecture! $\text{Det}[D[\{(1+xy)^3z + y^2(1+xy)(4+3xy), y + 3x(1+xy)^2z + 3xy^2(4+3xy), 2x - 3x^2y - x^3z\}, \{x,y,z\}]]$. It is remarkable that the jacobian is constant, that is an exceptional amount of cancellation. Does this polynomial map have any symmetry or other structure that makes this cancelation less miraculous?

Worked for 2m 54s >

Yes. The cancellation becomes quite transparent after introducing two auxiliary rational coordinates. Moreover, the map seems to possess a **three-sheeted/cubic structure**, rather than being an arbitrary collection of polynomials.

Write

$$F = (P, Q, R),$$

where

$$\begin{aligned} P &= (1 + xy)^3z + y^2(1 + xy)(4 + 3xy), \\ Q &= y + 3x(1 + xy)^2z + 3xy^2(4 + 3xy), \\ R &= 2x - 3x^2y - x^3z. \end{aligned}$$

1. A hidden auxiliary polynomial

Set

$$A = (1 + xy)^2z + y^2(4 + 3xy).$$

Then immediately

$$P = (1 + xy)A, \quad Q = y + 3xA, \quad R = x(2 - 3xy - x^2z).$$

Thus the first two coordinates are generated from the single polynomial A .

More revealingly, introduce

$$u = 1 + xy, \quad r = 2 - 3xy - x^2z.$$

Then

$$y = \frac{u-1}{x}, \quad z = \frac{5-3u-r}{x^2},$$

and a small calculation gives

$$x^2A = u + 1 - ru^2.$$

Consequently, in the birational coordinates (x, u, r) , the map is

$$\begin{aligned} P &= \frac{u(u+1-ru^2)}{x^2}, \\ Q &= \frac{4u+2-3ru^2}{x}, \\ R &= xr. \end{aligned} \tag{1}$$

The apparent denominators cancel when these expressions are pulled back to the original (x, y, z) -coordinates.

2. The Jacobian cancellation is essentially one line

The coordinate change

$$(x, y, z) \longmapsto (x, u, r)$$

has Jacobian

$$\det \frac{\partial(x, u, r)}{\partial(x, y, z)} = -x^3.$$

On the other hand, directly from (1),

$$\det \frac{\partial(P, Q, R)}{\partial(x, u, r)} = \frac{2}{x^3}.$$

Therefore

$$\det \frac{\partial(P, Q, R)}{\partial(x, y, z)} = \frac{2}{x^3}(-x^3) = -2.$$

So the cancellation is not a miraculous term-by-term coincidence. The map is constructed so that the pole x^{-3} in the Jacobian of the rational transformation exactly

cancels the x^3 in the birational coordinate change.

This resembles a cluster-type or birational-conjugation construction: one builds a simple rational étale map in adapted coordinates and arranges a Laurent-phenomenon-style cancellation making all three output coordinates polynomial.

3. A hidden cubic equation for the fibers

There is an even stronger indication of structure. Suppose $(P, Q, R) = (X, Y, Z)$. Eliminating u from (1) gives an equation for the original coordinate x :

$$D(X, Y, Z)x^3 + (4 - 3YZ)x - 2Z = 0, \quad (2)$$

where

$$D = 27X^2Z^2 - 18XYZ + 16X + Y^3Z - Y^2.$$

Thus the map appears generically to be a degree-three covering: the original x -coordinate is recovered as one of the roots of a depressed cubic.

There is a striking further identity:

$$(4 - 3YZ)^3 + 27DZ^2 = (27XZ^2 - 9YZ + 8)^2. \quad (3)$$

Hence the discriminant of the cubic (2) is

$$-4D(27XZ^2 - 9YZ + 8)^2.$$

The square factor is another unmistakable sign that the coefficients were engineered from a cubic-cover construction.

For example,

$$F(1, 1, 0) = (14, 22, -1).$$

For this target point, equation (2) becomes

$$-2(x - 1)(36x^2 + 36x + 1) = 0,$$

whose three distinct roots are

$$x = 1, \quad x = \frac{-3 + 2\sqrt{2}}{6}, \quad x = \frac{-3 - 2\sqrt{2}}{6}.$$

One would still substitute back to recover u, y, z , but the elimination calculation strongly suggests three distinct preimages and hence generic degree 3.

Interpretation

The essential structure is therefore:

$$(x, y, z) \xrightarrow{\text{birational}} (x, u, r) \xrightarrow{\text{simple Laurent map}} (P, Q, R),$$

with Jacobians

$$-x^3 \quad \text{and} \quad 2x^{-3}.$$

So:

- the constant Jacobian is structurally forced;
- the coefficients 3, 4, 2 encode the identities

$$x^2 A = u + 1 - ru^2$$

and (3);

- the map has a hidden cubic, apparently three-sheeted geometry;
- the right conceptual setting may be a rational self-map of a threefold, regularized by a Laurent cancellation.

I could not yet locate a public source for the purported announcement. Public references still describe the complex Jacobian conjecture as open, so the provenance and the argument establishing noninvertibility should be checked very carefully.

 Wikipedia +1 But the displayed polynomial itself is certainly not random: its constant Jacobian follows from the explicit birational factorization above.



Sources

Monday 11:01 PM

 Uploaded a file

After the first announcement of a solution, an alternate approach has been given in the attached PDF. Can you compare this approach with the previous example and see if it retroactively helps explain the construction of the former?

Worked for 26s >

Yes. I checked the identities independently, and the two maps are mutually inverse polynomial isomorphisms.

Let

$$X = \{(a, b, c, d, e) \in \mathbb{C}^5 : a^2e - abd + cb^2 = 1, \quad ad + bc = 1\}.$$

Define $\Phi : \mathbb{A}^3 \rightarrow X$ by

$$\begin{aligned} b &= 1 + ay, \\ c &= 1 - \frac{3}{2}ay + a^2z, \\ d &= \frac{1}{2}y - az + \frac{3}{2}ay^2 - a^2yz, \\ e &= -2z + 4y^2 - 4ayz + 3ay^3 - 2a^2y^2z. \end{aligned}$$

Direct expansion gives

$$ad + bc = 1$$

and

$$a^2e - abd + cb^2 = 1$$

identically in a, y, z . Thus Φ is defined on all of $\mathbb{A}^3$, including $a = 0$.

Now define $\Psi : X \rightarrow \mathbb{A}^3$ by

$$y = 2bd - ae,$$

$$z = 2d^2 + ce + 6bd^2 + 3bce - \frac{9}{2}e.$$

Substituting the forward formulas into these expressions gives identically

$$2bd - ae = y$$

and

$$2d^2 + ce + 6bd^2 + 3bce - \frac{9}{2}e = z.$$

Hence

$$\Psi \circ \Phi = \text{id}_{\mathbb{A}^3}.$$

Conversely, start with $(a, b, c, d, e) \in X$, define y, z by the polynomial formulas above, and reconstruct

$$b' = 1 + ay,$$

$$c' = 1 - \frac{3}{2}ay + a^2z,$$

$$d' = \frac{1}{2}y - az + \frac{3}{2}ay^2 - a^2yz,$$

$$e' = -2z + 4y^2 - 4ayz + 3ay^3 - 2a^2y^2z.$$

Reducing $b' - b, c' - c, d' - d, e' - e$ modulo the two defining equations of X gives zero in each case. Therefore

$$\Phi \circ \Psi = \text{id}_X.$$

So indeed

$$X \cong \mathbb{A}^3$$

by the explicit polynomial isomorphism

$$(a, y, z) \longleftrightarrow (a, b, c, d, e)$$

displayed above.

In particular, there is no hidden restriction or exceptional locus at $a = 0$; the apparent divisions in the preliminary formulas for d and e have removable singularities, and the final maps are polynomial everywhere.



If one composes Φ with the map
 $F(a,b,c,d,e) = (ac, ad + bc, ae + bd, be)$
 and then drops the $ad+bc=1$ term, can you confirm
 symbolically that one obtains a polynomial from (a,y,z) to
 $\mathbb{C}^3$ with constant Jacobian?

Yes. After composing Φ with

$$F(a, b, c, d, e) = (ac, ad + bc, ae + bd, be)$$

and discarding the identically constant coordinate $ad + bc = 1$, one gets the
 polynomial map

$$G : \mathbb{C}_{a,y,z}^3 \rightarrow \mathbb{C}^3$$

given by

$$G(a, y, z) = (G_1, G_2, G_3),$$

where

$$G_1 = ac = a - \frac{3}{2}a^2y + a^3z,$$

$$G_2 = ae + bd = \frac{1}{2}y - 3az + 6ay^2 - 6a^2yz + \frac{9}{2}a^2y^3 - 3a^3y^2z,$$

and

$$G_3 = be = -2z + 4y^2 - 6ayz + 7ay^3 - 6a^2y^2z + 3a^2y^4 - 2a^3y^3z.$$

Equivalently,

$$G_1 = a - \frac{3}{2}a^2y + a^3z,$$

$$G_2 = \frac{1}{2}y - 3az + 6ay^2 - 6a^2yz + \frac{9}{2}a^2y^3 - 3a^3y^2z,$$

$$G_3 = -2z + 4y^2 - 6ayz + 7ay^3 - 6a^2y^2z + 3a^2y^4 - 2a^3y^3z.$$

A direct symbolic differentiation gives

$$\det \frac{\partial(G_1, G_2, G_3)}{\partial(a, y, z)} = -1.$$

Thus G is a polynomial map with constant nonzero Jacobian.

A quick consistency check at $a = 0$ is

$$G(0, y, z) = \left(0, \frac{y}{2}, 4y^2 - 2z\right),$$

whose Jacobian matrix at $a = 0$ is triangular enough to give determinant -1 , consistent with the global calculation.

So this composition produces exactly the desired Keller map

$$G : \mathbb{C}^3 \rightarrow \mathbb{C}^3, \quad \det DG \equiv -1.$$



The original counterexample to the Jacobian conjecture is $F(z_1, z_2, z_3) = \left((1+z_1 z_2)^3 z_3 + z_2^2 (1+z_1 z_2) (4+3z_1 z_2), z_2 + 3 z_1 (1+z_1 z_2)^2 z_3 + 3 z_1 z_2^2 (4+3z_1 z_2), 2 z_1 - 3 z_1^2 z_2 - z_1^3 z_3 \right)$. Can you see how this relates to the map G ?

Worked for 22s >

Yes. They are the same map, up to a linear change of source coordinates, a reversal of the output coordinates, and two output rescalings.

Set

$$a = z_1, \quad y = z_2, \quad z = -\frac{1}{2}z_3.$$

Recall

$$G = (G_1, G_2, G_3),$$

where

$$G_1 = a - \frac{3}{2}a^2 y + a^3 z.$$

Under this substitution,

$$2G_1 = 2z_1 - 3z_1^2 z_2 - z_1^3 z_3,$$

which is exactly the third coordinate of the original counterexample.

Likewise, direct expansion gives

$$2G_2 = z_2 + 3z_1(1 + z_1z_2)^2z_3 + 3z_1z_2^2(4 + 3z_1z_2),$$

the second coordinate, and

$$G_3 = (1 + z_1z_2)^3z_3 + z_2^2(1 + z_1z_2)(4 + 3z_1z_2),$$

the first coordinate.

Consequently, writing F_{orig} for the displayed counterexample,

$$F_{\text{orig}}(z_1, z_2, z_3) = (G_3, 2G_2, 2G_1) \left(z_1, z_2, -\frac{z_3}{2} \right).$$

Equivalently, define the linear maps

$$A(z_1, z_2, z_3) = \left(z_1, z_2, -\frac{1}{2}z_3 \right)$$

and

$$B(u_1, u_2, u_3) = (u_3, 2u_2, 2u_1).$$

Then

$$F_{\text{orig}} = B \circ G \circ A.$$

This also checks the Jacobian constants. Since

$$\det A = -\frac{1}{2}, \quad \det B = -4, \quad \det DG = -1,$$

we obtain

$$\det DF_{\text{orig}} = (-4)(-1) \left(-\frac{1}{2} \right) = -2,$$

which agrees with the Jacobian of the original formula.

So the factorization construction has recovered the original map exactly, modulo these elementary linear normalizations.



